Binomial Coefficients

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1079 | Mathematics | https://cses.fi/problemset/task/1079

Your task is to calculate n binomial coefficients modulo 10^9+7.

A binomial coefficient a choose b can be calculated using the formula $(a!)/(b!(a-b)!)$. We assume that a and b are integers and $0 \le b \le a$.

Input

The first input line contains an integer n: the number of calculations.

After this, there are n lines, each of which contains two integers a and b.

Output

Print each binomial coefficient modulo 10^9+7.

Constraints

- $1 \le n \le 10^5$
- $0 \le b \le a \le 10^6$

Example

Input

```
3
5 3
8 1
9 5
```

Output

```
10
8
126
```